

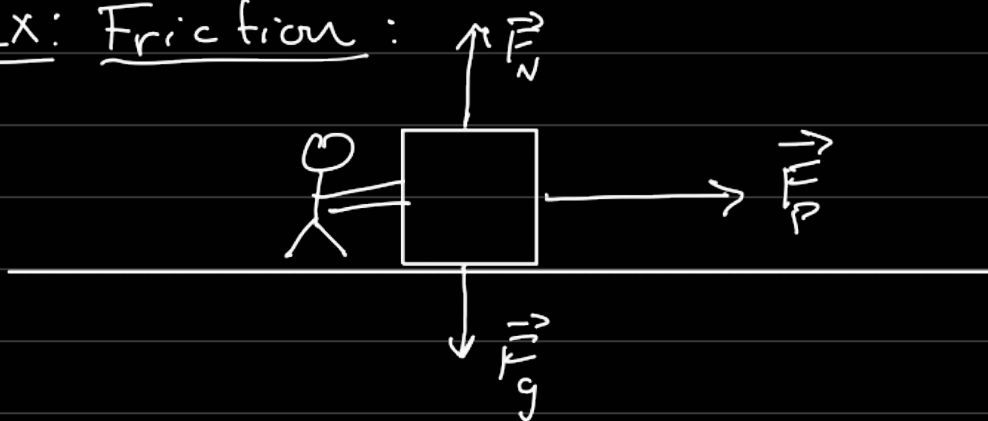
* Math $\leftrightarrow$ physics

• physics: The world we "see"

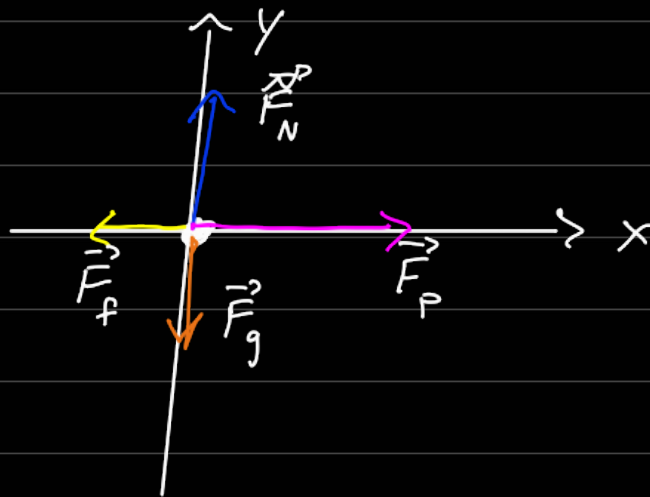
Math: Not "related" but is the language we use:

- we set up the problem in any way we'd like (math set up) but the physics should not depend on the set up.

ex: Friction:



- For this system we set up a coordinate system



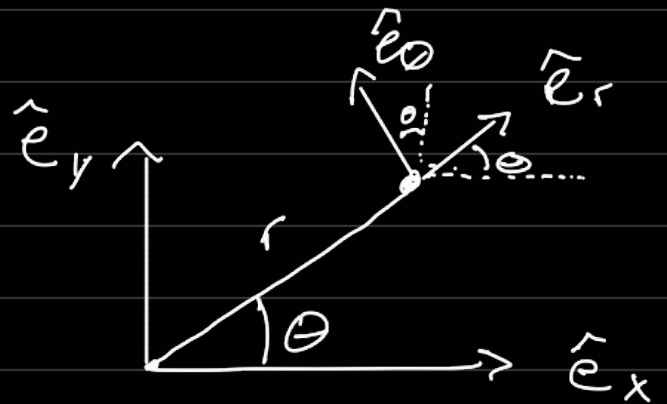
- we choose this coordinate system. Could have also chosen



• This makes things much more complicated. So, point of our coord. choice is to make the problem easier.

* Box on a slope:

Reference-Frames / Circular motion:



$$|\hat{e}_r| = 1$$

$$(\hat{e}_r) = \cos(\theta) \hat{e}_x + \sin(\theta) \hat{e}_y$$

$$(\hat{e}_\theta) = -\sin(\theta) \hat{e}_x + \cos(\theta) \hat{e}_y$$

$$V_{r,\theta} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} V_{x,y}$$

$$V_{x,y} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} V_{r,\theta}$$

$$\hat{e}_x = \cos(\theta) \hat{e}_r - \sin(\theta) \hat{e}_\theta$$

$$\hat{e}_y = \sin(\theta) \hat{e}_r + \cos(\theta) \hat{e}_\theta$$

$$\vec{r} = r \hat{e}_r = r \cos(\theta) \hat{e}_x$$

$$\frac{d\hat{e}_r}{dt} = \hat{e}_\theta \quad \bigg| \quad \frac{d\hat{e}_\theta}{dt} = -\hat{e}_r$$

$$\vec{r} = r \hat{e}_r$$

$$\begin{aligned} \vec{v} &= \frac{d\vec{r}}{dt} = \frac{d}{dt}(r \hat{e}_r) = \frac{dr}{dt} \hat{e}_r + r \frac{d\hat{e}_r}{dt} \\ &= \dot{r} \hat{e}_r + r \frac{d\hat{e}_r}{dt} \end{aligned}$$

$$\frac{d\hat{e}_r}{dt} = \frac{d\hat{e}_r}{d\theta} \frac{d\theta}{dt} = \dot{\theta} \hat{e}_\theta$$

$$\vec{v} = \dot{r} \hat{e}_r + r \dot{\theta} \hat{e}_\theta$$

same for $\vec{a}$

$$\vec{a} = (\ddot{r} - r \dot{\theta}^2) \hat{e}_r + (r \ddot{\theta} + 2\dot{r} \dot{\theta}) \hat{e}_\theta$$

$$r = a e^{k\varphi} = a e^{k\omega t} \quad \theta = \omega t$$

$$\ddot{r} = a (k\omega)^2 e^{k\omega t} = (k\omega)^2 r$$

$$\ddot{\theta} = \omega$$

$$\ddot{\theta} = 0$$

$$\dot{r} = k\omega r$$

$$\begin{aligned} \vec{a} &= (k^2 \omega^2 - \omega^2) r \hat{e}_r + 2 k \omega r \omega \hat{e}_\theta \\ &= \omega^2 (k^2 - 1) \vec{r} + 2 k \omega \dot{\vec{r}} \end{aligned}$$

$$F = m \vec{a} =$$

